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Batch: April 2026

E-Commerce Customer Churn Prediction and Retention System

ABSTRACT

This project presents an end-to-end Machine Learning (ML) system designed to predict customer churn, estimate Customer Lifetime Value (CLV), and provide root-cause interpretability for targeted retention strategies. Utilizing a synthetic dataset generated across 500 customers, the engine processes multi-channel transaction data, web session histories, and customer support ticket interactions. Features are structured via an automated Extract, Transform, Load (ETL) pipeline utilizing Recency, Frequency, and Monetary (RFM) behavioral profiles. To address demographic class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) is applied prior to training an eXtreme Gradient Boosting (XGBoost) classifier. Individualized customer churn drivers are extracted programmatically using SHAP (SHapley Additive exPlanations) Tree Explainers. Simultaneously, a Random Forest Regressor models a target proxy of customer financial health. Predictions are integrated into a secure, production-ready Streamlit web application dashboard restricted via an enterprise authentication layer. The deployment simulation demonstrates a high-impact financial recovery strategy, converting a projected 18% of identified high-risk accounts to preserve enterprise revenue channels.

INTRODUCTION

Retaining existing customers is significantly more cost-effective than acquiring new ones in highly competitive digital ecosystems. Modern web-centric

organizations produce massive pipelines of distributed operational data, including financial transactions, digital application browsing trails, and customer service records. However, raw digital exhaust is rarely unified, limiting an enterprise's capability to react to early behavioral warning signs of customer attrition.

This project delivers a complete enterprise engineering solution. It transitions raw logs into automated predictive models that provide clean tactical business indicators. The system merges predictive power with Explainable AI (XAI). This combination enables customer success teams to understand not only who is going to leave, but also why they are leaving.

PROBLEM STATEMENT

Enterprise analytical teams face three persistent structural bottlenecks when building retention tools:

- **Data Silos:** Behavioral indicators are fragmented across decoupled operational repositories (transaction databases, web application telemetry logs, and CRM support platforms).
- **Class Imbalance:** Churned users represent a minority group within overall historical files. Standard model architectures biased toward active majorities fail to identify true churn signals accurately.
- **Black-Box Metrics:** Predictive probability scores without supporting context fail to suggest structural business solutions. This lack of transparency causes marketing teams to miss actionable tactical points.

The project objectives are to construct a programmatic data engine, map individual risk matrices, account for class imbalances, and deliver a protected interactive interface to address these exact limitations.

RESULTS AND DISCUSSION

1. Pipeline Synthesis & Model Selection

The backend system initializes structural dependencies (xgboost, shap, lifetimes, streamlit, plotly, imbalanced-learn) and generates testing profiles for 500 unique corporate customers. The programmatic pipeline constructs an ETL framework that extracts:

- RFM Core Behavioral Metrics: Recency values, transaction repetition frequencies, and historical monetary spend aggregates.
- Web Telemetry Aggregates: Average session duration periods, cumulative product page views, and shopping cart abandonment metrics.
- CRM Service Metrics: Support ticket frequencies, resolution speed windows, and numeric customer sentiment indexes mapped from categorical logs.

2. Analytical Matrix Findings

The structured enterprise dataset yields a multi-dimensional matrix of size (500, 11). Churn classification labels are established by tracking users with more than 30 days of consecutive transaction inactivity. After applying SMOTE to balance the training set, the XGBoost engine creates definitive risk groups.

Risk Stratification Matrix Layout:

	Low Risk (< 70% Churn Prob)	High Risk (>= 70% Churn Prob)
High Value (CLV)	High Value, Low Risk Quadrant (Core Retained Revenue)	High Value, High Risk Quadrant (Immediate Executive Priority)
Low Value (CLV)	Low Value, Low Risk Quadrant (Stable Self-Service Accounts)	Low Value, High Risk Quadrant (Low Priority Maintenance)

Individual root-cause extraction via SHAP AI identifies specific points of friction. It details precise reasons for churn, such as elevated support resolution times, recurrent cart abandonments, or dropping transactional frequency rates.

3. Dashboard Implementation & Economic Savings

The client web layer is written as a specialized local production file (app.py) featuring an enterprise gatekeeper key ("april2026"). The interactive workspace provides top-line system indicators, cross-sectional segmentation views, and granular individual log look-ups.

An embedded financial simulation tracks users with high churn probabilities (Probability ≥ 0.70). By routing these flagged accounts to targeted customer retention workflows, the business simulates an 18% conversion rate. The system calculates projected retained revenue dynamically by multiplying converted accounts by their predicted Random Forest CLV. This calculation provides an executive view of measurable cost savings directly on the dashboard.



5. CONCLUSION

This project demonstrates that integrating machine learning classifiers with explainable AI models creates highly effective proactive customer retention programs. The automated framework successfully consolidates disconnected database logs into an actionable, analysis-ready feature matrix. Overcoming structural minority imbalance through SMOTE ensures stable classification performance. Using SHAP analytics removes the mystery from model predictions, allowing operations teams to quickly identify the root causes of customer dissatisfaction. Deploying these models through a secure Streamlit web application gives enterprises real-time control over revenue risk management.

6. REFERENCES

- Documentation on Data Scaling & Tree Boosting: XGBoost Developer Documentation Engine (<https://readthedocs.io>)
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- Documentation on Imbalanced Class Handling (SMOTE): Imbalanced-Learn PyPI Integration Manual (<https://imbalanced-learn.org>)
- Documentation on Interface Design & Web Caching Platforms: Streamlit Application Workspace Architecture Guide (<https://streamlit.io>)